

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

Register Number	192324237
Name	D.Manjunadha

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

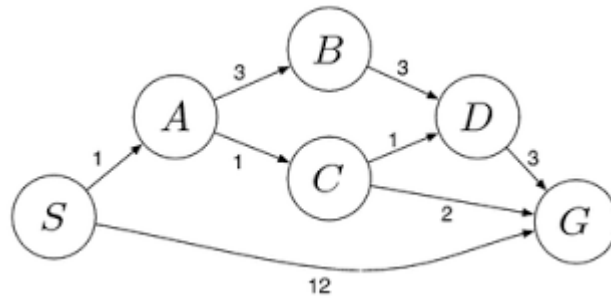
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path

from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

*I D.Manjunadha(192324237)(Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: D.Manjunadha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

1Q)

Represent the amount of water in the 4-gallon jug and 3-gallon jug as 4-gallon, 3-gallon.

Step	Action	State (4-gal, 3-gal)
1	Fill the 3-gallon jug	(0, 3)
2	Pour the 3-gallon jug into the 4-gallon jug	(3, 0)
3	Fill the 3-gallon jug again	(3, 3)
4	Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full (only 1 gallon is needed)	(4, 2)
5	Empty the 4-gallon jug onto the ground	(0, 2)
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	(2, 0)

The 4-gallon jug now contains exactly 2 gallons of water, which is the required solution

Q2)

i. What are the percepts for this agent?

Percepts are the information the Mars Rover receives through its sensors.

- Images from navigation and scientific cameras
- Rock and soil composition data from spectrometers
- Temperature, pressure, and weather readings
- Terrain and obstacle information

ii. Characterize the operating environment

The Mars Rover operates in the following environment:

- Partially Observable: The rover cannot observe the entire environment at once.
- Sequential: Current actions affect future actions.
- Dynamic: Weather conditions and terrain can change.
- Continuous: Position, movement, and time are continuous.

iii. What are the actions the agent can take?

- Move forward, backward, left, or right
- Avoid obstacles
- Capture images and videos
- Drill rocks and collect soil samples
- Analyze samples using scientific instruments

iv. How can one evaluate the performance of the agent?

The Mars Rover's performance can be evaluated based on:

- Accuracy of scientific observations
- Number and quality of samples collected
- Successful navigation without collisions
- Completion of assigned missions
- Efficient use of battery power
- Reliability of data transmitted to Earth

v. What sort of agent architecture is most suitable for this agent? Why?

Reason:

- Chooses actions that maximize mission success while minimizing risks.
- Learns from previous experiences to improve navigation and decision-making.
- Balances multiple objectives such as safety, power conservation, and scientific value.
- Can adapt to unknown terrain and changing environmental conditions.
- Makes autonomous decisions because communication with Earth has significant delays.

3Q)

The 8-Queens problem is to place 8 queens on an 8×8 chessboard so that no two queens attack each other.

A queen can attack horizontally, vertically, and diagonally.

1. Initial State

- An empty 8×8 chessboard with no queens placed.

2. State Space

- Each state represents a configuration of queens on the board.
- A common representation is one queen per column, placing queens column by column.
- Total possible configurations (without constraints): $8^8 = 16,777,216$.
- With one queen per row and one per column: $8! = 40,320$ configurations.

3. Goal Test

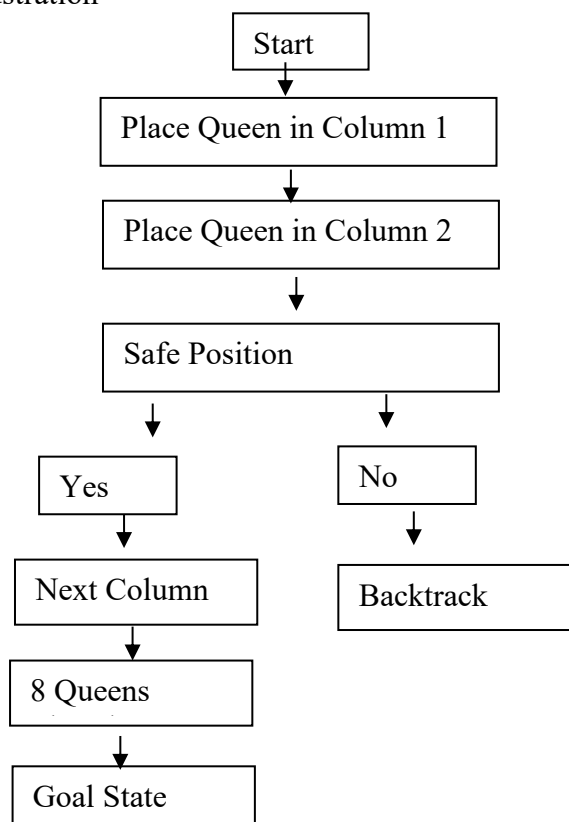
A solution is reached when:

- All 8 queens are placed, and
- No two queens share:
 - The Same row,
 - the same column,
 - the same diagonal.

4. Path Cost

- Each queen placement has a cost of 1.
- Since all solutions require placing 8 queens, the path cost is generally not important.

State-Space Search Illustration



4Q)

Algorithm:

1. START
2. Open OLA Application
3. Read Current_Location
4. Enter Destination
5. Get Available_Cabs
6. (Mini, Micro, Sedan, Shared, Prime, etc.)
7. Display Fare and Estimated Arrival Time
8. Customer selects preferred cab type
9. Confirm Booking
10. Assign nearest available driver
11. Driver picks up customer
12. Navigate to destination
13. Complete trip
14. Calculate fare
15. Make payment
16. Display "Ride Completed Successfully"
17. .STOP

Initial State

- Customer opens the OLA app.
- Current location is detected.

Actions

- Enter destination.
- Display available cab types.
- Compare fare and ETA.
- Select preferred cab.
- Confirm booking.
- Track the ride until the destination.

Goal State

- Customer reaches the destination safely using the selected cab.

Performance Measure

- Minimum travel time.
- Lowest fare.
- Maximum comfort.
- Successful ride completion.
- Customer satisfaction.

5Q)

Given weighted graph:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Uniform Cost Search (UCS)

Step Node Expanded Path Cost Frontier (Priority Queue)

1	S	0	A(1), G(12)
2	A	1	C(2), B(4), G(12)
3	C	2	D(3), G(4), B(4), G(12)
4	D	3	G(4), B(4), G(6), G(12)
5	G	4	Goal Found

Cost Calculation

Path 1

$S \rightarrow G$

Cost = 12

Path 2

$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$

Cost = $1 + 3 + 3 + 3 = 10$

Path 3

$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$

Cost = $1 + 1 + 1 + 3 = 6$

Path 4 $S \rightarrow A \rightarrow C \rightarrow G$

Cost = $1 + 1 + 2 = 4$

Final Answer

- Least-Cost Path: $S \rightarrow A \rightarrow C \rightarrow G$
- Total Cost: 4